

Vidyarthi-Bus

Crowdsourced Bus Alert System

Android App Development using Generative AI

Project Requirements & Specification Document

Version 1.0	Domain EdTech / Rural Transit	Platform Android (Kotlin)	Backend Firebase + GenAI
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1. Problem Statement

Students in remote villages depend entirely on specific college buses for daily commute. When a bus is full or cancelled, students miss exams and lectures with no alternative information readily available. The core gap is the absence of **real-time crowd visibility** — students at upcoming stops have no way of knowing whether there is room on the bus before it arrives.

Key Pain Points

- No visibility into bus occupancy before it reaches the next stop.
- Last-minute scramble for alternatives leads to missed exams.
- Rural areas have poor cellular coverage, demanding a lightweight solution.
- No centralised channel for bus cancellation or delay alerts.

2. Project Vision & Description

Vidyarthi-Bus is a **crowdsourced bus-alert Android app** that empowers students to report and receive real-time bus crowd status. Students already on the bus report the crowd level with a single tap; students waiting at upcoming stops immediately see a colour-coded indicator and can decide whether to wait or seek an alternative ride.

Mission Statement

"Eliminate transport uncertainty for rural students by turning every passenger into a real-time data point — making every bus ride predictable."

3. Scope of the Application

Dimension	In Scope	Out of Scope
Users	College students (reporters & viewers)	Bus drivers, college admin
Platform	Android (API 26+)	iOS, Web, Desktop
Routes	Pre-defined college bus routes	City buses, trains, rideshare
Data	Crowd status, GPS proximity check, shared-auto contacts	Ticketing, payments, live GPS tracking of bus
Backend	Firebase Realtime DB, Firebase Auth	Custom server, ML model training
GenAI Usage	AI-assisted code generation, smart alt-transport suggestions	Fully autonomous AI decision-making

4. App Usage & User Flow

Step	Action	Screen / Component
1	Open app & authenticate via college email	Login Screen (Firebase Auth)
2	Select bus route number from dropdown	Route Selection Screen
3	View Crowd Meter — colour-coded status bar	Dashboard (Green/Yellow/Red bar)
4	Tap 'I'm on the bus' to report crowd level	One-tap Report Button
5	System validates reporter's GPS proximity to route	Background Location Service
6	Status broadcasts to all users on same route instantly	Firebase Realtime DB Listener
7	If bus is full, view alternate Shared Auto contacts	Alternatives Screen

5. Functional Requirements

ID	Requirement	Description	Priority
FR-01	User Authentication	Login with college email via Firebase Auth; block non-college accounts.	HIGH
FR-02	Route Selection	Display all registered bus routes; user selects one to view its crowd status.	HIGH
FR-03	Real-time Crowd Meter	Horizontal progress bar changes colour: Green (0-40%), Yellow (41-70%), Red (71-100%). Updates in under 2 s.	HIGH
FR-04	One-tap Report	Single button lets a student report their presence on the bus. Report expires automatically after 15 minutes.	HIGH
FR-05	Location Proximity Check	App verifies reporter is within 500 m of the bus route before accepting a report. Prevents fraudulent data.	HIGH
FR-06	Report Timeout / Expiry	Firebase Cloud Function auto-deletes stale reports older than 15 minutes to keep crowd data fresh.	HIGH
FR-07	Alternatives Listing	When bus is full (Red), display a list of nearby shared-auto contact numbers for the route.	MEDIUM
FR-08	Push Notifications	Send FCM notification to route subscribers when status changes from Green to Red.	MEDIUM
FR-09	Bus Cancellation Alert	Designated route captain can post a 'Bus Cancelled' alert visible to all route subscribers.	MEDIUM

6. Non-Functional Requirements

ID	Category	Requirement	Target
NFR-1	Performance	Crowd status update latency across all connected clients	< 2 seconds
NFR-2	Performance	App cold-start time on a mid-range Android device	< 3 seconds
NFR-3	Availability	System uptime during college hours (6 AM – 9 PM)	99.5 %
NFR-4	Scalability	Concurrent users supported per route without degradation	≥ 200 users
NFR-5	Security	Firebase Security Rules must block unauthenticated read/write access to crowd data	Zero public writes
NFR-6	Data Integrity	Stale reports must not persist beyond the timeout window	15-min TTL
NFR-7	Usability	New user can report crowd status without any prior training (discoverability)	≤ 3 taps

NFR-8	Accessibility	UI must be readable in bright sunlight; minimum contrast ratio for text	WCAG AA (4.5:1)
NFR-9	Network	App must degrade gracefully on 2G / low-bandwidth connections; show cached last status	Offline-aware
NFR-10	Privacy	Individual location data must never be stored; only a proximity boolean is used	No PII stored

7. Feature Catalogue

7.1 Must-Have Features (MVP)

Feature	Description	Linked FR
College Email Login	Restrict app access to verified college students only via Firebase Authentication.	FR-01
Route Selection	Scrollable list of bus routes. Tapping a route navigates to its live crowd dashboard.	FR-02
Live Crowd Meter	Horizontal progress bar with three zones: Green (seats available), Yellow (filling up), Red (full). Updates live from Firebase.	FR-03
One-tap Crowd Report	A prominent 'I'm on the bus' button. Adds the reporter to the active count in Firebase.	FR-04
GPS Proximity Gate	Validate that the reporter's device GPS is within 500 m of the bus route polyline before accepting the report.	FR-05
Auto-Expiry of Reports	Cloud Function scheduled trigger removes crowd entries older than 15 minutes.	FR-06
Shared Auto Contacts	Static list of trusted alternate transport contacts shown when the bus is full.	FR-07

7.2 Good-to-Have Features (Post-MVP)

Feature	Description	Value Add
Push Notifications	FCM alerts when a subscribed route status changes to Red or bus is cancelled.	Proactive awareness
Bus Cancellation Alert	Route captain can post a cancellation; all subscribers notified immediately.	Reduces wasted wait time
AI Smart Suggestions (GenAI)	Use Gemini API to suggest the best alternate route or departure time based on historical crowd patterns.	GenAI differentiator

Crowd History Chart	Show a weekly bar chart of average crowd levels per route per time slot.	Helps plan travel time
Dark Mode	Full Material 3 dark theme for early-morning low-light usage.	UX comfort
Multi-language Support	Localisation for regional languages (Kannada, Telugu, Tamil, Hindi).	Rural inclusivity
Gamification / Leaderboard	Reward frequent accurate reporters with points to encourage data quality.	Data quality incentive
Offline Mode	Cache last known crowd status locally; show a 'Last updated X min ago' banner when offline.	Works on 2G

8. Technical Implementation

Layer	Technology	Purpose
Language	Kotlin (Android SDK 26+)	Primary development language
UI Framework	Jetpack Compose / XML + Material 3	Crowd Meter as HorizontalProgressBar
Real-time DB	Firebase Realtime Database	Sub-second crowd status sync
Auth	Firebase Authentication (Google/Email)	College identity verification
Location	Google Fused Location Provider API	GPS proximity check for reporters
Cloud Functions	Firebase Cloud Functions (Node.js)	15-min report TTL, notification triggers
Notifications	Firebase Cloud Messaging (FCM)	Push alerts on status change
GenAI Layer	Google Gemini API / Android Studio Gemini	Code generation, smart suggestions

9. Impact Goals

■ Educational Equity	■ Smart Transit	■ Reliability
Ensure transport hurdles never prevent a rural student from reaching an exam or class.	Use community-sourced micro-data to solve infrastructure gaps without expensive hardware.	Make rural public transport predictable through real-time, crowd-validated information.

10. Success Criteria

- The Crowd Meter updates for ALL users on a route within 2 seconds of a new report.
- The app prevents users who are NOT within 500 m of the bus route from submitting a report.
- The UI cold-starts in under 3 seconds on a mid-range device with 2 GB RAM.
- Stale reports are automatically purged within 15 minutes — the meter reflects only live data.
- At least 3 successful crowd reports are made per bus trip during pilot testing.
- User satisfaction score $\geq 4/5$ in post-pilot usability survey among rural students.